1. Aim

To identify the major classes in the Digital Health Management System, classify them into strong and weak classes based on their dependency and existence, and represent their relationships using a UML Class Diagram.

2. Introduction

The Digital Health Management System (DHMS) is designed to manage healthcare operations digitally. It provides a centralized platform for patient registration, appointment scheduling, medical records, consultations, prescriptions, laboratory reports, and notifications.

In Object-Oriented Software Engineering, a **class** represents a group of objects having common attributes (data) and behaviours (operations or methods).

Strong class: A class that has an independent identity and can exist independently of another class in the model.

Weak class: A class whose existence or lifecycle depends on another class. It usually represents a dependent record or component.

3. Identification of Classes

S.No.	Class Name	Description
1	**User**	Stores user account and authentication details.
2	**Role**	Defines user roles and permissions.
3	**Patient**	Stores patient personal and medical information.
4	**Clinician**	Stores doctor or healthcare professional details.
5	**Appointment**	Manages patient appointments and scheduling.
6	**Encounter**	Stores details of a patient consultation or visit.
7	**Prescription**	Stores prescriptions issued by clinicians.
8	**PrescriptionItem**	Stores individual medicines and dosage instructions.
9	**Medication**	Stores medicine information.
10	**LabOrder**	Stores laboratory test orders.
11	**LabReport**	Stores diagnostic results and report details.
12	**Notification**	Manages appointment and follow-up notifications.
13	**AuditEvent**	Maintains records of important system activities.

4. Classification of Strong and Weak Classes

A. Strong Classes

Class	Reason
User	Represents an individual user account.
Role	Defines system permissions independently of a particular user.
Patient	Represents an individual patient and their medical profile.
Appointment	Has its own booking ID and lifecycle.
Medication	Represents a medicine in the medication catalogue.

B. Weak Classes

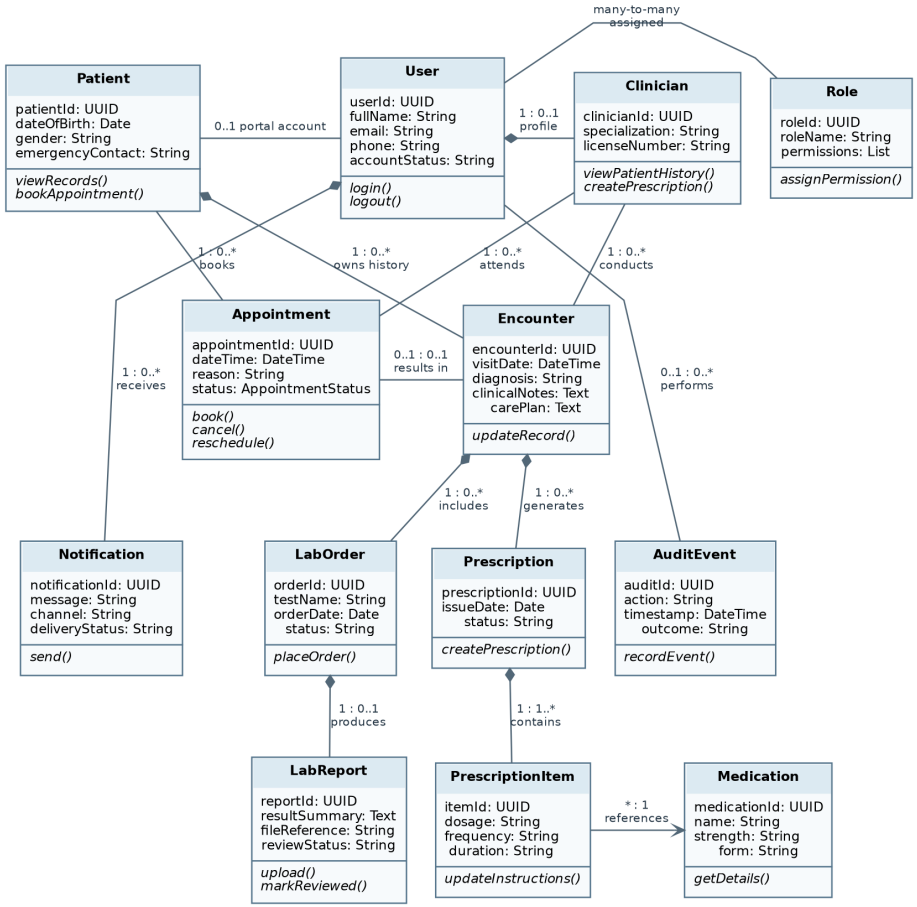
Class	Dependent On	Reason
Clinician	User	A clinician profile is linked to a user account.
Encounter	Patient	A consultation belongs to a patient's medical history.
Prescription	Encounter	A prescription is generated during a clinical encounter.
PrescriptionItem	Prescription	Individual medicine entries belong to a prescription.
LabOrder	Encounter	A laboratory order is associated with a patient's clinical visit.
LabReport	LabOrder	A laboratory report is generated against a test order.
Notification	User / triggering event	A notification requires a recipient and triggering event.
AuditEvent	User / system event	An audit event records an action performed by an actor or system.

Classification note: “Weak” refers to lifecycle dependency in this conceptual model, not poor design. Some dependent classes may still have their own unique ID in the implemented database.

5. UML Class Diagram

The diagram shows the principal classes, selected attributes and operations, associations, multiplicities, and composition relationships. Filled diamonds indicate composition: the dependent part is treated as belonging to the lifecycle of its owning class.

DIGITAL HEALTH MANAGEMENT SYSTEM - UML CLASS DIAGRAM



Multiplicity guide: 1 = exactly one; 0..1 = optional; 0..* = zero or many; 1..* = one or many.

6. Explanation of Relationships

Relationship	Explanation
User - Role	A user can have multiple roles, and each role can be assigned to multiple users.
Patient - Appointment	One patient can book multiple appointments, but each appointment belongs to one patient.
Clinician - Appointment	One clinician can handle multiple appointments.
Patient - Encounter	A patient can have multiple clinical encounters throughout their medical history.
Encounter - Prescription	A clinical encounter can generate multiple prescriptions.
Prescription - PrescriptionItem	A prescription contains one or more medicine entries.
PrescriptionItem - Medication	Each prescription item references a medication.
Encounter - LabOrder	A clinical encounter can include multiple laboratory test orders.
LabOrder - LabReport	A laboratory order can produce a diagnostic report.
User - Notification	A user can receive multiple notifications.
User - AuditEvent	A user can perform multiple actions that are recorded in the audit log.

7. Conclusion

In this laboratory exercise, 13 important classes were identified for the Digital Health Management System and classified into strong and weak classes. Strong classes maintain independent identities, while weak classes depend on other classes for their lifecycle.

The UML Class Diagram illustrates the associations, multiplicities, and dependencies among the classes. It provides a structural view of the DHMS and forms a foundation for object-oriented design and implementation.

Design note: Patient and User are modelled separately because a patient record may exist even when the patient does not have a portal account. A Clinician profile is linked to a User account. This keeps healthcare records distinct from login credentials.

End of Lab 3